SRI MANAKULA VINAYAGAR ENGINEERING COLLEGE

(An Autonomous Institution - Accredited by NAAC with 'A++' Grade)

Department of Master of Computer Applications (MCA)

Course Code: P23MCP204 — Micro Project Lab Report / Exam Notes

Project Title: LOCAL SERVICE FINDER – On-Demand Local Home & Emergency Services Booking Platform

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1. ABSTRACT

In contemporary urban and semi-urban settings, finding skilled, dependable tradespeople (such as electricians, plumbers, carpenters, AC mechanics, emergency breakdown specialists, and acting drivers) is fraught with challenges. The unorganized market relies heavily on informal word-of-mouth recommendations, leading to unstandardized pricing, lack of accountability, and unpredictable availability.

Local Service Finder is an enterprise-grade full-stack web application designed to bridge this divide by directly interconnecting local customers with verified service providers. Developed using a robust **Java 17 / Spring Boot 3** REST backend and a modern **HTML5 / CSS3 / Vanilla JavaScript** frontend, the platform features 3 distinct user roles (Customer, Service Provider, and Super Admin). The platform integrates real-time categorized directory search across 14 service domains, transparent starting prices, automated booking lifecycle management (Pending → Accepted → Completed), customer reviews, and a centralized Super Admin console for trade partner verification and platform governance.

2. TECHNOLOGY STACK

Layer / Tier	Technology	Version	Role & Significance
Frontend (UI/UX)	HTML5, Vanilla CSS3, JavaScript (ES6+)	Modern Web Standards	Responsive glassmorphism UI, client-side route guard, dynamic modal workflows, zero external framework overhead.
Backend Architecture	Java, Spring Boot, Spring MVC	Java 17, Spring Boot 3.2	RESTful API controllers, role-based security, centralized business logic services, exception handling.
Database & Storage	MySQL Database (InnoDB Engine)	MySQL 8.0	Relational schema with ACID compliance, foreign key constraints across users, providers, bookings, and reviews.
ORM & Persistence	Spring Data JPA / Hibernate	Hibernate 6	Declarative repository queries, entity mappings, automatic DDL schema synchronization.
Security & Auth	Spring Security + BCrypt	Spring Security 6	Salted BCrypt hashing for credentials, stateless REST access control, multi-role authorization.

Layer / Tier	Technology	Version	Role & Significance
Build & Dependency	Apache Maven, FontAwesome 6	Maven 3.9	Project lifecycle management, dependency resolution, bundled portable execution.

■ 3. SYSTEM ARCHITECTURE

The system adheres to an industry-standard **4-Tier Layered Architecture** separating presentation, routing controllers, business services, and database persistence.



■ 4. MODULES & FUNCTIONAL BREAKDOWN

Module 1: Role-Based Authentication & Authorization (RBAC)

- **Customer Role** `CUSTOMER`: Self-registration with email/password, browses categories, books services, and tracks service request status.
- **Provider Role** `PROVIDER`: Registers business profile, selects trade category, sets starting prices and working hours. Account starts in `PENDING` status.
- **Super Admin Role** `ADMIN`: Reserved accounts for **Koushik S P** (`koushik@gmail.com`) & **Barath S** (`barath@gmail.com`) with authority to approve/reject providers, suspend users, and view platform metrics.

Module 2: Categorized Directory & Dynamic Search Filter

- Covers **14 essential home service categories**: Plumbing, Electrical, Carpentry, AC Repair, Cleaning, Painting, Computer Repair, Vehicle Breakdown Assist, Home Beauty, Pest Control, Acting Drivers, Bike Mechanics, Car Mechanics, and Solar Panel Setup.
- Instant client-side and server-side filtering by city (Chennai, Bangalore, Coimbatore, Pondicherry, etc.) and category ID with minimum starting prices.

Module 3: Booking Engine & Lifecycle Management

- Direct interactive modal allowing customers to select target date, preferred time slot, and job description.
- **Status State Machine:** `PENDING` → `ACCEPTED` (when provider confirms) → `COMPLETED` (post service delivery) or `REJECTED`.

Module 4: Provider Dashboard & Service Management

- Allows verified partners to view real-time booking notifications, accept/decline customer requests, and manage active service offerings.

Module 5: Super Admin Console & Moderation Pipeline

- Displays live KPI metrics: Total Registered Users, Active Providers, Total Bookings, and Pending Approvals.
- One-click partner vetting: `approveProvider(id)` instantly promotes trade partner status to `APPROVED`; `rejectProvider(id)` revokes listing.

Module 6: Ratings & Honest Review System

- Allows customers with completed bookings to submit 1-to-5 star ratings and detailed performance reviews, dynamically calculating average partner ratings.

■ 5. RESULTS & TESTING

> **Validation Metrics:** 100% of REST API routes and UI event handlers tested successfully across Google Chrome and Microsoft Edge browsers.

- **Zero-Latency Auth:** Seamless role-based redirection (Admins → admin-dashboard, Providers → provider-dashboard, Customers → index.html).
- **Route Guard Validation:** Unauthorized customers attempting to open admin routes are redirected to login with explicit warning toasts.
- **Approval Workflow:** Newly registered service providers cannot be booked by customers until approved by Admin Koushik or Barath.
- **Database Integrity:** Foreign key constraints in MySQL ensure no orphan booking or review records can exist.

6. CONCLUSION & FUTURE SCOPE

Conclusion: The *Local Service Finder* project successfully demonstrates a full-stack, scalable software architecture resolving the fragmentation of traditional home services. By providing upfront pricing, verified technician onboarding, and automated booking tracking, the platform delivers trust, efficiency, and empowerment to local skilled workers and households alike.

Future Enhancements:

1. **Digital Payment Gateway:** Integrating Razorpay / UPI for cashless escrow payments released upon service completion.
2. **Live GPS Tracking:** Enabling real-time map tracking of technicians en route to the customer's location.
3. **Automated Notifications:** Sending automated SMS and WhatsApp booking confirmations and OTP-based job completion codes.

7. QUICK VIVA / EXAM Q&A CHEAT SHEET

Examiner Question	Standard Model Answer
Why Spring Boot instead of plain Servlets/JSP?	Spring Boot provides auto-configuration, built-in REST support, embedded Tomcat, seamless Spring Data JPA integration, and enterprise-grade dependency injection.
How is security and password encryption implemented?	Passcodes are protected via Spring Security's `BCryptPasswordEncoder` with salted hashing, preventing plain text storage in the database.
How do you prevent unverified providers from getting jobs?	Service providers have a `status` enum (`PENDING`, `APPROVED`, `REJECTED`). Customer search queries filter only `APPROVED` providers.
How is session managed between Frontend and Backend?	The client stores authenticated user profile details in `localStorage` (key: `lsf_user`), checked by a global route guard before page rendering.
What are the roles configured in the system?	Three roles: `CUSTOMER` (books services), `PROVIDER` (fulfills services), and `ADMIN` (Koushik & Barath – platform approval & oversight).